

Problem 1:

Let X be a locally compact Hausdorff space. Prove that X is second countable if and only if its one-point compactification is second countable.

Proof.

($\Leftarrow$) Assume the one-point compactification of X is second countable. Thus, it has a countable basis β .

Since the topology of the one point compactification is made up of open sets of X and sets of the form $\{X \cup \{\infty\}\} \setminus K$ s.t. K is compact in X .

Therefore, we know that for any $B \in \beta$ that $B \cap X$ is open. Since, if $B \subset X$ then it is already open in X .

if not then $B = (X \cup \{\infty\}) \setminus K$. Thus, $B \cap X = X \setminus K$ which is open in X since K is compact thus closed. Thus, $X \setminus K$ is open.

Thus, consider $\beta_X = \{B \cap X \mid B \in \beta\}$

Let $x \in X$. Since X is a subset of the one point compactification this means that $x \in B$ for some $B \in \beta$. Thus, $x \in B \cap X \in \beta_X$.

If $x \in B_1 \cap B_2$ for some $B_1, B_2 \in \beta_X$. Then we know that $B_1 = U_1 \cap X$ and $B_2 = U_2 \cap X$ for $U_1, U_2 \in \beta$.

Thus, $x \in U_1 \cap U_2$. Thus, there exists a $U_3 \in \beta$ s.t. $x \in U_3 \subset U_1 \cap U_2$.

Thus, $x \in U_3 \cap X \subset B_1 \cap B_2$.

Thus, β_X is a countable basis for X .

($\Rightarrow$) Assume X is second countable.

Let β be a countable basis for X .

Define $\beta' = \{B \in \beta \mid \text{there exists a compact } K_B \subset X \text{ with } B \subset K_B\}$.

Let $x \in X$. Since X is locally compact we know there is a neighborhood U of x s.t. $U \subset K$ for K being compact.

Since β is a basis we know there is a B s.t. $x \in B \subset U \subset K$. Thus, β' covers X .

Let $x \in B_1 \cap B_2$ for some $B_1, B_2 \in \beta'$. Thus, there is a K_1, K_2 s.t. these are compact and $B_1 \subset K_1$ and $B_2 \subset K_2$. Since K_1, K_2 are compact this implies they are closed. Thus, $K_1 \cap K_2$ is closed. Since X is Hausdorff this means $K_1 \cap K_2$ is compact.

Thus, $x \in B_1 \cap B_2 \subset K_1 \cap K_2$. Since β , we know there exists a $B_3 \in \beta$ s.t. $x \in B_3 \subset B_1 \cap B_2 \subset K_1 \cap K_2$ thus $B_3 \in \beta'$.

Thus, β' is a basis for X .

Define $F = \{U \subset X \mid U = \bigcup_{i=1}^m B_i, B_i \in \beta', m \in \mathbb{N}\}$ which is countable since β' is countable.

Thus, for each set $U = B_1 \cup B_2 \cdots \cup B_n \in F$ we know for each $B_i \subset K_i$ s.t. K_i is compact.

Thus, each K_i is closed. Thus, the finite union of all the K_i 's is also closed, and since X is Hausdorff this means the union is also compact. Call this union K_U .

Let $\hat{X}$ be the one point compactification of X .

Define $V_U = \hat{X} \setminus K_U$

Let $\hat{B} = \beta' \cup \{V_U \mid U \in F\}$

Let $x \in \hat{X}$

If $x \in X$ then $x \in B$ for some $B \in \beta'$

If $x = \infty$ then let V be an open neighborhood of ∞ . By definition of the topology on $\hat{X}$ we know that $V = \hat{X} \setminus K$ s.t. K is compact.

Since β' is an open cover of K we know there exists a finite subcover $B_1, \dots, B_n \in \beta'$ s.t. $K \subset B_1 \cup \dots \cup B_n$.

Thus, $K \subset K_1 \cup \dots \cup K_n = K_U$. Since each $B_i \subset K_i$ for some K_i that is compact in X .

Thus, $\infty \in V_U \subset V$.

Let $U_1, U_2 \in \hat{B}$ s.t. $x \in U_1 \cap U_2$.

If $x \in X$ then since open sets in $\hat{X}$ are either already open in X or of the form $\hat{X} \setminus K$ where K is compact. This means that the open sets intersected with X are also open in X . We do not have to check for the first type but for the second type when intersecting with X it becomes $X \setminus K$. Since K is compact it is closed thus $X \setminus K$ is open.

Therefore, any open set in $\hat{X}$ intersected with X is open.

Thus, $x \in U_1 \cap U_2 \cap X$. Since β' is a basis this implies there exists a $B \in \beta'$ s.t. $x \in B \subset U_1 \cap U_2 \cap X$. Thus, $x \in B \subset U_1 \cap U_2$. Thus, done since $B \in \beta'$.

If $x = \infty$ then $U_1 = \hat{X} \setminus K_{F_1}$ and $U_2 = \hat{X} \setminus K_{F_2}$. Thus, $K = K_{F_1} \cup K_{F_2}$ is also compact in X .

Thus, by a similar proof above by showing K is a subset of basis elements in β' and taking a new set $K \subset K_U$. We know that $\infty \in V_U \subset \hat{X} \setminus K = U_1 \cap U_2$.

Thus, $\hat{B}$ is a basis for $\hat{X}$ which is countable. ■

Problem 2:

For each space below, determine which of the countability axioms (first countable, second countable, and separable) X satisfies.

(a)

$X = \mathbb{R}^\omega$, equipped with the topology induced by the uniform metric.

Proof.

Let $x \in \mathbb{R}^\omega$ with the uniform metric then consider $\{B(x, 1/n) \mid n \in \mathbb{N}\}$.

Let U be an open neighborhood of x . Then, there exists an $\epsilon > 0$ s.t. $B(x, \epsilon) \subset U$.

Fix $N \in \mathbb{N}$ s.t. $1/N < \epsilon$. Thus, $B(x, 1/N) \subset B(x, \epsilon) \subset U$.

Thus, $\mathbb{R}^\omega$ is first countable.

Assume $\mathbb{R}^\omega$ has a countable dense subset U .

Consider $f : \{0, 1\}^\omega \rightarrow U$ s.t. $f(x) = u$ s.t. $u \in B(x, 1/2)$.

Injective:

Let $x, y \in \{0, 1\}^\omega$ s.t. $f(x) = f(y)$. Thus, $x = y$ since $B(x, 1/2)$ only contains one point from $\{0, 1\}^\omega$.

Thus, this implies U is uncountable thus, $\mathbb{R}^\omega$ is not separable.

Thus, since $\mathbb{R}^\omega$ is not separable proven in class this implies it is not second countable. ■

(b)

X is the set $\mathbb{R}$ equipped with the topology generated by the basis of intervals of the form $[a, b)$, for $a, b \in \mathbb{R}$.

Proof.

Let $x \in X$ then consider $\{[x, x + 1/n) \mid n \in \mathbb{N}\}$.

Let U be an open neighborhood of x . Then, there exists an interval s.t. $x \in [a, b) \subset U$.

Thus pick $N \in \mathbb{N}$ s.t. $a + 1/N < b$. Thus, $x \in [a, a + 1/N) \subset U$.

Thus, X is first countable.

Consider $\mathbb{Q}$. $\mathbb{Q}$ is dense since any set $[a, b)$ will have a rational number in it. This is from $\mathbb{Q}$ being dense in $\mathbb{R}$ with standard topology. Thus, any open set intersects $\mathbb{Q}$ implying that $\overline{\mathbb{Q}} = \mathbb{R}$.

Thus, X is separable.

Assume for contradiction that X is second countable. Thus, it has a second countable basis β .

Define $f : X \rightarrow \beta$ s.t. $f(x) = B_x$ s.t. $x \in B_x \subset [x, x+1)$.

Injective:

Let $x, y \in X$ s.t. $f(x) = f(y)$. Thus, $B_x = B_y$. Thus, $B_y = B_x \subset [x, x+1)$ and $B_y = B_x \subset [y, y+1)$.

Thus, $x = y$.

This is a contradiction since X is uncountable thus X is not second countable. ■

Problem 3:

Show that a compact metrizable space is always second countable. (This is exercise 4 in §30 of Munkres; see that exercise if you need a hint.)

Proof.

Let X be compact and metrizable. Let d be a metric on X .

Consider $\mathcal{C} = \{B(x, 1/n) \mid x \in X\}$. This is an open cover of X . Since X is compact we know there is a finite subcover B_n .

Consider $\beta = \bigcup_{n \in \mathbb{N}} B_n$

Let $x \in X$ since each B_n is a finite subcover $x \in B \in B_n$.

Let $x \in U_1 \cap U_2$ s.t. $U_1, U_2 \in \beta$. Thus, there is an $\epsilon > 0$ s.t. $x \in B(x, \epsilon) \subset U_1 \cap U_2$.

Now consider B_N s.t. $1/N < \epsilon/2$. Since B_N is a finite subcover $x \in B$ for some $B = B(y, 1/N) \in B_N$.

Thus, for $z \in B$ we have $d(x, z) \leq d(x, y) + d(y, z) \leq 2/N < \epsilon$. Thus, $x \in B(y, 1/N) \subset U_1 \cap U_2$.

Thus, β is a countable basis for X . Thus, X is second countable. ■

Problem 4:

Let X_1, X_2 be two spaces consisting of the same underlying set X equipped with topologies $\mathcal{T}_1, \mathcal{T}_2$ respectively; suppose that $\mathcal{T}_1$ is finer than $\mathcal{T}_2$.

(a)

What (if anything) can you conclude about X_1 if X_2 is assumed to be Hausdorff?
What (if anything) can you conclude about X_2 if X_1 is assumed to be Hausdorff?

Proof.

Assume X_2 is Hausdorff. Let $x, y \in X_1$. Since $\mathcal{T}_2 \subset \mathcal{T}_1$. We know in $\mathcal{T}_2$ there exists disjoint nonempty open sets s.t. $x \in U$ and $y \in V$.

Since $U, V \in \mathcal{T}_2 \subset \mathcal{T}_1$. Thus, X_1 is Hausdorff.

The other way does not work. Consider $\mathcal{T}_1 = \mathcal{P}(X)$, $\mathcal{T}_2 = \{\emptyset, X\}$, and $X = \mathbb{R}$. X_1 is Hausdorff but X_2 is not. ■

(b)

Repeat part (a), but replace “Hausdorff” with “regular” and “normal” in turn.

Proof.

Both ways do not work for regular. For one direction use the same counter example as above.

For the other direction

Consider $\mathbb{R} = X_2$ with standard topology and $X_1 = \mathbb{R}$ with a basis of (a, b) and $(a, b) \setminus K$ s.t. $K = \{1/n \mid n \in \mathbb{N}\}$.

Suppose there exists disjoint open sets containing $0, K$. Called U, V . Let B be a basis element containing 0 that is a subset of U . It must be a basis element of the form $(a, b) \setminus K$ since for any open interval containing zero intersects K .

Fix $n \in \mathbb{N}$ large enough s.t. $1/n \in (a, b)$. Now pick a basis element about $1/n$ contained in V . It must be an interval (c, d) since $1/n \in K$.

This is a contradiction since fix $z \in \mathbb{R}$ s.t. $\max c, 1/(n+1) < z < 1/n$. Thus, $z \in U \cap V$ which is a contradiction.

Thus, X_1 is not regular but X_2 is.

Both ways do not work for normal. For one direction use the same counter example as above.

For the other direction

We can also use the example of $\mathbb{R}$ with standard topology and $\mathbb{R}$ with the special K

■ topology. Since it is not regular it is also not normal, but $\mathbb{R}$ is normal. ■

Problem 5:

Prove that if M_1 is an n -manifold and M_2 is an m -manifold, then the space $M_1 \times M_2$ is an $(n + m)$ -manifold.

Proof.

Let $U \times V$ be a basic open set in $M_1 \times M_2$. The reason we only have to look at basic open sets is because every set is a union of these so when applying a function $f(\cup B) = \cup f(B)$.

Thus, U is open in M_1 and V is open in M_2 .

Thus, $U \cong S$ and $V \cong T$ s.t. S is an open set in $\mathbb{R}^n$ and T is an open set in $\mathbb{R}^m$.

Let f be a homeomorphism for U to S and let g be the same for V, T .

Consider, $h : U \times V \rightarrow S \times T$ s.t. $h(x, y) = (f(x), g(y))$. Thus, this is a homeomorphism since f, g are homeomorphisms. Thus, $U \times V \cong S \times T \subset \mathbb{R}^{n+m}$.

Therefore, $M_1 \times M_2$ is a $n + m$ manifold. ■

Problem 6:

Consider the cube $C = [-1,1] \times [-1,1] \times [-1,1]$. This cube has six faces, given by the subsets

$$\begin{aligned} &\{\pm 1\} \times [-1,1] \times [-1,1], \\ &[-1,1] \times \{\pm 1\} \times [-1,1], \\ &[-1,1] \times [-1,1] \times \{\pm 1\}. \end{aligned}$$

We can form the space T_3 by taking the quotient of C by “identifying opposite faces;” precisely, by taking $T_3 = C / \sim$, where $x \sim y$ if either $x = y$, or if any of the following hold:

$$\begin{aligned} x = (\pm 1, u, v), & \quad y = (\mp 1, u, v) \\ x = (u, \pm 1, v), & \quad y = (u, \mp 1, v) \\ x = (u, v, \pm 1), & \quad y = (u, v, \mp 1) \end{aligned}$$

Prove that T_3 is homeomorphic to the product $S^1 \times S^1 \times S^1$, and therefore a manifold. *Hint:* it suffices to find a continuous bijection from T_3 to the product; why? You may also want to apply Problem 5 on Homework 4 (Theorem 22.2 in Munkres).

Proof.

Let p be the quotient map.

Define $g : C \rightarrow S^1 \times S^1 \times S^1$ s.t. $g(x, y, z) = (e^{i\pi x}, e^{i\pi y}, e^{i\pi z})$.

Since $e^{i\pi x}$ is continuous we know that g is continuous.

Let $x \neq y \in C$ s.t. $x \sim y$.

Thus, let $x = (\pm 1, u, v)$ and $y = (\mp 1, u, v)$. However, since $e^{-i\pi} = e^{i\pi}$. This, implies that $g(x) = g(y)$. The same argument works for the other types of equivalences.

Thus, g is constant on the equivalence classes. Therefore, by Theorem 22.2 we get an induced map $f : T_3 \rightarrow S^1 \times S^1 \times S^1$ s.t. $f \circ p = g$.

Also, since g is continuous we know that f is continuous.

Injective:

Let $a, b \in T_3$ s.t. $f(a) = f(b)$.

Since p is surjective this implies there is $x, y \in C$ s.t. $p(x) = a$ and $p(y) = b$.

Thus, $f(p(x)) = f(p(y))$. Therefore, $g(x) = g(y)$.

Thus, $(e^{i\pi x_1}, e^{i\pi x_2}, e^{i\pi x_3}) = (e^{i\pi y_1}, e^{i\pi y_2}, e^{i\pi y_3})$

Thus, $e^{i\pi(x_j - y_j)} = 1$

This, implies that $\pi(x_j - y_j) = 2\pi k$ for some $k \in \mathbb{Z}$.

Dividing by π gets:

$$x_j - y_j = 2k$$

Since $x_j, y_j \in [-1, 1]$. This implies that k is 0, 1, or -1 .

Therefore this implies that either:

$$\begin{aligned} x_j - y_j = 0 &\implies x_j = y_j \\ x_j - y_j = 2 &\implies x_j = 1, y_j = -1 \\ x_j - y_j = -2 &\implies x_j = -1, y_j = 1 \end{aligned}$$

Thus, this implies that $x \sim y$.

Therefore, $a = b$. Thus, f is injective.

Surjective:

Let $x \in S^1 \times S^1 \times S^1$. Therefore, since $e^{i\pi t} : [-1, 1] \rightarrow S^1$ is surjective there is x_1, x_2, x_3 s.t. $x = (e^{i\pi x_1}, e^{i\pi x_2}, e^{i\pi x_3})$.

Thus, consider $f(q(x_1, x_2, x_3)) = g(x_1, x_2, x_3) = x$.

Therefore, f is surjective.

First, since C is closed and bounded we know that it is compact. Thus, $q(C) = T_3$ is compact. Also, since $S^1 \times S^1 \times S^1$ is a subspace of a Hausdorff space it is Hausdorff. Thus, since f is a continuous bijection from a compact space to a Hausdorff space it is a homeomorphism. Therefore, $T_3 \cong S^1 \times S^1 \times S^1$. ■